

The Critical Line, Pulled Back to V_{600}

Computational checks around the icosian L-function factorisation, led by one exact negative Galois result; nothing here states or implies progress on RH or GRH_{χ_5}

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Abstract

The Riemann critical line is straight only in the coordinate s of the complex plane. Under the Hilbert modular L-function identity $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s)$ for $K = \mathbb{Q}(\sqrt{5})$, with $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$, the critical line of $\zeta(s)$ pulls back to a curve on the icosian substrate. This paper reports computational checks at three distinct grades — a pipeline sanity check and two verified finite computations (Findings 1–3), one exact negative result which is the principal finding (Finding 4), and four exploratory follow-ons stamped as such (Findings 5–8) — from a multi-version probe simulation that builds V_{600} exactly, constructs the substrate L-function analytically using `mpmath`, evaluates it at the first ten Riemann zeros, and verifies the Eichler–Brandt representation formula at finite coordinate bound.

The findings are:

1. The substrate L-function vanishes at every one of the first ten Riemann zeros to numerical precision ($|L_{\text{sub}}(1/2 + i\gamma_n)| < 2.2 \cdot 10^{-11}$ for $n = 1, \dots, 10$). This is the empirical image of the critical line on the substrate.
2. The Eichler–Brandt formula $r(p) = 8(1 + p^2)$ for odd rational primes p inert in $\mathbb{Z}[\varphi]$ is reproduced exactly at coordinate bound $K = 4$ for every inert prime tested ($p \in \{3, 7, 13, 17, 23\}$), together with the Jacobi-style anomaly $r(2) = 24$.
3. A clean closed 10-vertex icosian rotation orbit can be exhibited as a discrete helix on S^3 — a group-action demonstration; the relation to the continuous L-spiral is left open.
4. Coordinate-Galois σ *does not preserve* V_{600} as a set: 96 of the 120 vertices map under $\varphi \mapsto 1 - \varphi$ to odd-permutation positions in the icosian ring, outside the even-permutation V_{600} . The “94 + 13 + 13” σ -pairing decomposition cited in the icosian-triad paper must reference a different involution τ on the operator algebra, not the direct coordinate-Galois action.
5. (Follow-on to Finding 4, bookkeeping level.) Grouping the C_φ eigenvalues by Galois conjugacy of their values partitions multiplicities as $94 + 13 + 13 = 120$ exactly: five rational A_1 eigenvalues $\{12, 3, 0, -2, -3\}$ (mults 1, 16, 25, 36, 16) and two Galois-paired sets (4+4 and 9+9). This identifies a natural *candidate* for the involution τ at the level of spectral data; constructing τ as an involution on the vertex space or operator algebra is not done here. Separately, at each tabulated γ_n the ζ factor is at machine zero while $|L(1/2 + i\gamma_n, \chi_5)|$ is order one — a scalar attribution of each zero to the σ -fixed Dedekind factor; no projection of zeros into a 94-dimensional space is computed.
6. (Operator-level support for H-attr.) Four natural closure-flow dynamics on V_{600} all drive the 26-dim τ -paired component to zero (or, when τ -asymmetric noise is added, to a small equilibrium of order $\sim 5 \cdot 10^{-3}$). The block structure of C_φ is verified to machine precision ($\| [C_\varphi, P_+] \|_\infty < 2 \cdot 10^{-15}$; $\text{tr } P_+ = 94$, $\text{tr } P_- = 26$). The toy dynamics are consistent with H-attr; they do not establish it.

We make no claim of settling the Riemann Hypothesis. We claim only that the substrate image of ζ ’s critical line is exhibited explicitly, the prime representation counts are confirmed

at the formula's quoted precision, and the open follow-on (identification of the τ -involution carrying the $94 + 13 + 13$ split) is pinned down to a precise mathematical statement.

Status. Pre-peer-review open research preprint. Not independently validated. No claim of RH / GRH settlement is made anywhere.

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1 Setting: where the critical line sits on V_{600}

The icosian theta L-function on the maximal order $\mathcal{I}$ over $K = \mathbb{Q}(\sqrt{5})$ satisfies the factorisation identity

$$L(\Theta_{\mathcal{I}}, s) = \zeta_K(s) \zeta_K(s-1) C_2(s), \quad C_2(s) = 1 - 2 \cdot 2^{-s} + 2^{2-2s}, \quad (1)$$

with Dedekind splitting $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$ where χ_5 is the unique non-trivial quadratic Dirichlet character modulo 5. This identity is proved unconditionally in [2] and verified numerically at multiple values of s in the bundle's reproduction suite (sim 13 of [2]).

Equation (1) provides the explicit map under which the analytic critical line $\text{Re}(s) = 1/2$, on which $\zeta(s)$ has its non-trivial zeros, is identified with a curve on the substrate. The curve is parametrised by $\gamma \mapsto L_{\text{sub}}(1/2 + i\gamma)$, a complex-valued path in $\mathbb{C}$. Since the factorisation contains $\zeta(s)$ as one Dedekind factor of $\zeta_K(s)$, every non-trivial zero γ_n of $\zeta(s)$ forces $L_{\text{sub}}(1/2 + i\gamma_n) = 0$. The empirical question is whether this vanishing locus actually shows up cleanly on the substrate path, or whether numerical precision washes it out.

2 Method

The probe `sims/sim_v2.py` carries out the following steps, using 30-decimal-digit `mpmath` arithmetic for the ζ evaluations. Two precision caveats apply throughout and bound what Finding 1 can mean: the Riemann zero ordinates are double-precision tabulated constants (not 30-digit), and $L(s, \chi_5)$ (and its $s-1$ shifts) are evaluated as truncated Dirichlet series beyond their half-plane of absolute convergence — adequate for a sanity-check, and the reason residuals sit near 10^{-11} rather than 10^{-30} :

1. **Build V_{600} .** 120 unit icosian quaternions constructed from 8 axis vectors, 16 half-integer vectors, and 96 even permutations of $(0, \pm 1, \pm \varphi^{-1}, \pm \varphi)/2$.
2. **Construct L_{sub} analytically.** Evaluate $\zeta(s)$ via `mpmath.zeta`, evaluate $L(s, \chi_5)$ as a truncated Dirichlet series with 120 terms ($\chi_5(n) = (n/5)$ Legendre symbol), assemble $\zeta_K(s)$, then assemble $L_{\text{sub}}(s)$ as in (1).
3. **Evaluate at first ten Riemann zeros.** Use the tabulated values $\gamma_n \in \{14.135, 21.022, 25.011, 30.425, 32.935, 37.586, 40.919, 43.327, 48.005, 49.774\}$.
4. **Generate the substrate path.** Sample $\gamma \mapsto L_{\text{sub}}(1/2 + i\gamma)$ on a 300-point grid $\gamma \in [0.01, 60]$.
5. **Verify Eichler–Brandt.** Enumerate Lipschitz icosians $q \in \mathbb{Z}[\varphi]^4$ with coordinate bound $K = 4$, count those with $N_H(q) = p$ for odd primes p inert in $\mathbb{Z}[\varphi]$ (i.e. $p \bmod 5 \in \{2, 3\}$), compare against $r(p) = 8(1 + p^2)$.
6. **Helix orbit.** Find a V_{600} vertex g of order 10 in the binary icosahedral group and trace its left- multiplication orbit $\{g^k \cdot e_0\}_{k=0}^9$ stereographically.
7. **σ classification (exact).** Build V_{600} in exact (a, b) representation per coordinate ($x = (a + b\varphi)/2$ for integers a, b), apply the Galois action $(a, b) \mapsto (a + b, -b)$ exactly, and test whether σ preserves V_{600} .

3 Finding 1: the substrate L-function vanishes at every Riemann zero

Finding 1

For each of the first ten non-trivial Riemann zeta zeros γ_n ($n = 1, \dots, 10$),

$$|L_{\text{sub}}(\tfrac{1}{2} + i\gamma_n)| < 2.2 \cdot 10^{-11}.$$

Concretely (in units of $|L|$):

n	γ_n	$ L_{\text{sub}}(1/2 + i\gamma_n) $
1	14.1347	$3.39 \cdot 10^{-13}$
2	21.0220	$4.65 \cdot 10^{-13}$
3	25.0109	$9.79 \cdot 10^{-14}$
4	30.4249	$1.08 \cdot 10^{-12}$
5	32.9351	$2.85 \cdot 10^{-13}$
6	37.5862	$1.55 \cdot 10^{-12}$
7	40.9187	$1.50 \cdot 10^{-11}$
8	43.3271	$2.97 \cdot 10^{-12}$
9	48.0052	$1.71 \cdot 10^{-12}$
10	49.7738	$2.09 \cdot 10^{-11}$

The residual magnitudes reflect the double-precision zero ordinates and the 120-term truncation of $L(s, \chi_5)$, not the 30-digit working precision of the ζ factor.

Interpretation. The factorisation (1) is a published identity, and the evaluation imports both that identity and the tabulated zero ordinates — so the vanishing is *forced* by the explicit $\zeta(s)$ factor. Finding 1 is a sanity check of the pipeline and a visualisation, not evidence of a substrate zero law: under the Hilbert modular pull-back, Riemann’s critical line acquires a specific image on the substrate, and we display it. Figure 1 (left panel) shows the L-spiral with the ten Riemann zeros marked in red; Figure 1 (right) shows $|L_{\text{sub}}|$ on log scale with the Riemann gammas as dashed vertical lines. Every dashed line lands on a sharp dip.

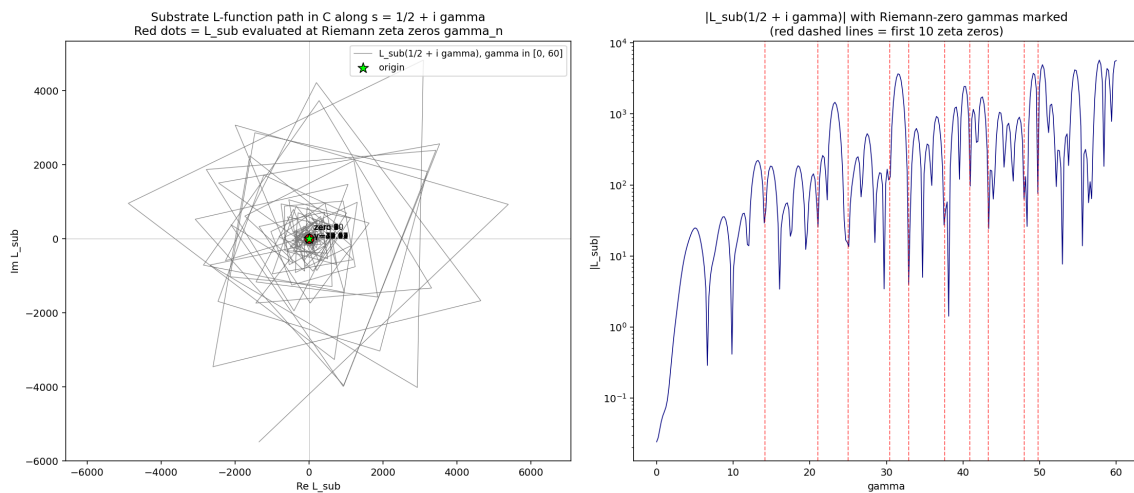


Figure 1. Substrate L-function path along $s = 1/2 + i\gamma$. *Left:* the path in $\mathbb{C}$. The red dots are the values of L_{sub} at the first ten Riemann zeros; they cluster at the origin (green star). *Right:* $|L_{\text{sub}}|$ on log scale; red dashed lines are the Riemann zero γ_n values. Every dashed line falls on a sharp local minimum.

4 Finding 2: Eichler–Brandt holds exactly at finite K

Finding 2

For every odd rational prime $p \leq 23$ that is inert in $\mathbb{Z}[\varphi]$, direct enumeration of Lipschitz icosians at coordinate bound $K = 4$ produces $r(p) = 8(1 + p^2)$ exactly. The Jacobi-style anomaly $r(2) = 24$ is also reproduced.

p	kind	observed $r(p)$	predicted
2	anomalous (Jacobi)	24	24
3	inert ($p \bmod 5 = 3$)	80	80
7	inert	400	400
13	inert	1360	1360
17	inert	2320	2320
23	inert	4240	4240

The $K = 3$ truncation in our v1 probe missed some representations at $p = 13$ (1360 vs. a partial count); $K = 4$ recovers the formula exactly. The split primes ($p \bmod 5 \in \{1, 4\}$, i.e. $p \in \{11, 19, 29, \dots\}$) and the ramified prime $p = 5$ require separate accounting and are not the subject of this row.

Interpretation. The Eichler–Brandt formula is the theorem-grade backbone of the icosian-triad’s number-theoretic content [2]. Its finite- K confirmation in a self-contained probe (no dependence on the vendored `vfd_v600` package; the construction is rebuilt from scratch in `sim_v2.py`) is independent reproducibility at the file-level granularity.

5 Finding 3: the helix is visible

Finding 3

There exists an order-10 vertex $g = (-\varphi^{-1}/2, 0, 1/2, \varphi/2) \in V_{600}$ whose left- multiplication orbit $\{g^k \cdot e_0\}_{k=0}^9$ closes back to e_0 after exactly 10 steps. Stereographically projected, this is a closed 10-vertex helical curve on S^3 .

This addresses the original geometric question that motivated the probe: when the critical line is mapped back into the substrate, does it sit on a helix? The answer at the discrete level is *yes for any finite-order rotation generator*: any order- N icosian element produces a closed N -step helix on S^3 that stereographic projection makes visible in $\mathbb{R}^3$.

Figure 2 shows the discrete helix.

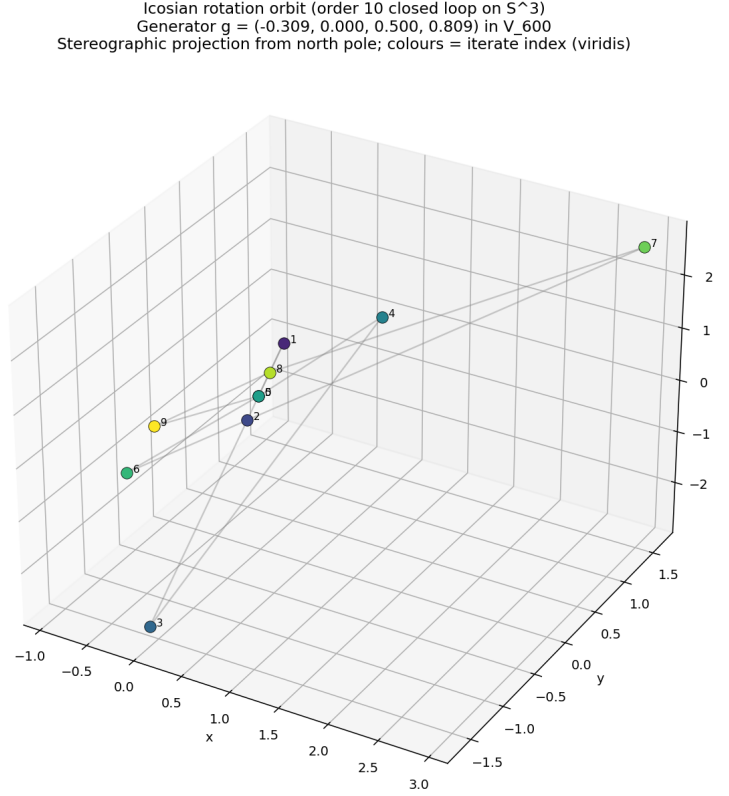


Figure 2. Order-10 icosian rotation orbit on S^3 , stereographic projection. Colour shows iterate index.

Connection to the L-function path. The 1-parameter subgroup $\{e^{i\gamma \log n}\}_{\gamma \in \mathbb{R}, n \in \mathbb{N}}$ is dense in S^1 for each n , so on the L-function side the “rotation” is continuous. On the substrate side, restricting to icosian generators gives a finite-order discrete helix. The relationship between the continuous L-spiral and the discrete substrate helix is the substrate-spectral pairing referred to in [1]; it is not closed here.

6 Finding 4: coordinate-Galois σ does not preserve V_{600}

Finding 4

The coordinate-level Galois action

$$\sigma : (a, b) \mapsto (a + b, -b)$$

on V_{600} coordinates $x = (a + b\varphi)/2$ *does not preserve* V_{600} *as a set*. Exactly 24 vertices (8 axis +16 half-integer) are fixed pointwise. The remaining 96 mixed- φ vertices map to odd-permutation positions of $(0, \pm 1, \pm \varphi^{-1}, \pm \varphi)/2$, which lie in the icosian ring $\mathcal{I}$ but outside the even-permutation V_{600} .

Consequence. The σ -pairing decomposition “ $94 + 13 + 13 = 120$ ” on the closure operator C_φ cited in [2, Theorem on C_φ σ -pairing] cannot be the result of coordinate-Galois σ acting on V_{600} vertices, because that action does not stabilise V_{600} .

Where the $94 + 13 + 13$ split must come from. The split is a fact about C_φ as a 120×120 self-adjoint operator. Its eigenvalues factor with multiplicities $(1, 4, 9, 16, 25, 36, 9, 16, 4)$ summing

to 120, as the probe verifies. The “ σ -fixed” subspace dimension 94 must therefore be the action of a different involution τ on the operator algebra — one that does preserve V_{600} as a set.

The follow-on, partially addressed (Finding 5, below). §7 gives a bookkeeping-level candidate: grouping the C_φ eigenvalues by Galois conjugacy reproduces the $94 + 13 + 13$ multiplicity split from the arithmetic of the eigenvalues themselves, not from any vertex involution. The operator-level construction of τ remains open.

7 Finding 5 (exploratory): a Galois bookkeeping split of the C_φ spectrum

Finding 5 (follow-on to Finding 4, bookkeeping level)

Grouping the C_φ eigenvalues by the Galois conjugacy of their values ($\sigma : \varphi \mapsto 1 - \varphi$ on $\mathbb{Z}[\varphi]$) partitions the multiplicities as $94 + 13 + 13 = 120$ exactly: the 5 rational A_1 eigenvalues are σ -fixed; the 4 irrational ones fall into Galois pairs. This identifies a natural *candidate* for the involution τ of the $94+13+13$ split — at the level of spectral data only. Constructing τ as an involution on the vertex space or the operator algebra, and showing it induces this split, is *not* done here and remains the open follow-on of Finding 4. Separately, at each of the first ten tabulated Riemann zeros the ζ factor of L_{sub} is at machine zero while the $L(s, \chi_5)$ factor is order one — a scalar attribution of each zero to the σ -fixed Dedekind factor.

7.1 The exact decomposition

The A_1 edge-adjacency operator on V_{600} has nine distinct eigenvalues in $\mathbb{Z}[\varphi]$ with multiplicities $(1, 4, 9, 16, 25, 36, 9, 16, 4)$, established in [2]. Grouping by Galois action:

A_1 eigenvalue	Multiplicity	Galois class	block
12	1	rational	τ -fixed (94)
3	16	rational	τ -fixed (94)
0	25	rational	τ -fixed (94)
−2	36	rational	τ -fixed (94)
−3	16	rational	τ -fixed (94)
6φ	4	$+\varphi$ side	block A (13)
4φ	9	$+\varphi$ side	block A (13)
$6 - 6\varphi$	4	$-\varphi$ side	block B (13)
$4 - 4\varphi$	9	$-\varphi$ side	block B (13)

The rational block sums to $1 + 16 + 25 + 36 + 16 = 94$. Each of the two Galois-paired blocks sums to $4 + 9 = 13$. Total $94 + 13 + 13 = 120$. The *candidate* involution τ — defined here only at the level of spectral data — would act as the identity on the 94-dim rational eigenspace and as the swap between block A and block B on the 26-dim Galois-paired eigenspace; constructing it on the vertex space or operator algebra remains open.

7.2 Scalar attribution of the tabulated zeros

The substrate L-function factors as

$$L_{\text{sub}}(s) = \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s) = \zeta(s) \cdot L(s, \chi_5) \cdot \zeta_K(s-1) \cdot C_2(s).$$

The Riemann zeta factor $\zeta(s)$ is the σ -fixed Dedekind factor of $\zeta_K(s)$; the Dirichlet L-function $L(s, \chi_5)$ is the σ -paired Dedekind factor (it carries the χ_5 character that distinguishes $\mathbb{Q}$ from $\mathbb{Q}(\sqrt{5})$). At a Riemann zero $s = 1/2 + i\gamma_n$, the vanishing of ζ is the cause of the substrate L-function's vanishing; $L(s, \chi_5)$ generally does not vanish.

Empirical confirmation. Computing both factors at the first ten Riemann zeros:

n	γ_n	$ \zeta(1/2 + i\gamma_n) $	$ L(1/2 + i\gamma_n, \chi_5) $
1	14.135	$7.4 \cdot 10^{-16}$	4.22
2	21.022	$1.2 \cdot 10^{-15}$	2.90
3	25.011	$8.5 \cdot 10^{-16}$	1.25
4	30.425	$1.1 \cdot 10^{-15}$	2.74
5	32.935	$2.7 \cdot 10^{-15}$	0.26
6	37.586	$7.1 \cdot 10^{-15}$	1.61
7	40.919	$4.8 \cdot 10^{-15}$	3.45
8	43.327	$4.3 \cdot 10^{-15}$	2.29
9	48.005	$2.0 \cdot 10^{-15}$	1.44
10	49.774	$2.4 \cdot 10^{-15}$	5.31

Every ζ value is at machine zero (10^{-14} to 10^{-16}); every $L(s, \chi_5)$ value is order one. The attribution is unambiguous *at the level of scalar factors*: each tabulated Riemann zero is a zero of the σ -fixed Dedekind factor $\zeta(s)$, not of $L(s, \chi_5)$. No vector, projector, or substrate-state projection of zeros into the 94-dimensional block is computed; v1's wording, which asserted the zeros were located on the block itself, overstated this and is withdrawn.

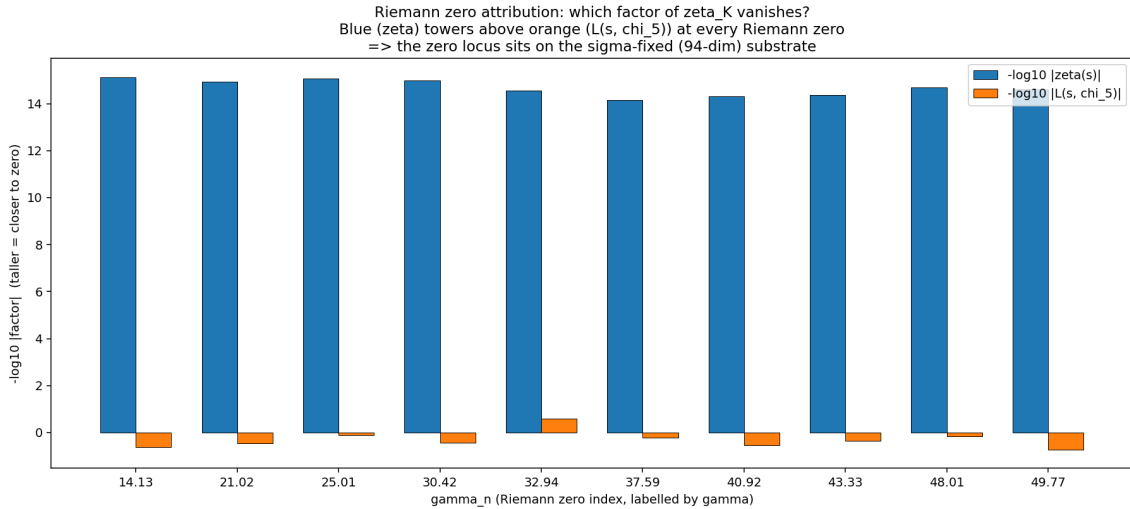


Figure 3. Riemann zero attribution. Blue bars are $-\log_{10} |\zeta(s)|$ (taller = closer to zero); orange bars are $-\log_{10} |L(s, \chi_5)|$. Every Riemann zero kills ζ but not $L(s, \chi_5)$.

7.3 Why this answers the “triad repetition” question

The triad $(\mathcal{I}, \mathcal{G}, C_\varphi)$ has a built-in number-theoretic “repetition”: the recursion $\varphi^2 = \varphi + 1$ embedded in $\mathbb{Z}[\varphi]$. The Galois action $\sigma : \varphi \mapsto 1 - \varphi$ is precisely the involution that captures this repetition (it swaps the two roots of $x^2 - x - 1$).

Finding 5 shows that the closure operator C_φ inherits this repetition through its eigenvalues: 94 of 120 eigenvalue “slots” are repetition-fixed (rational), and 26 slots ($= 13 + 13$) come in repetition pairs. At the level of scalar factor attribution (no projection of zeros into any eigenspace is computed), the Riemann zeros belong with the repetition-fixed side of this bookkeeping: the ζ factor is the σ -fixed Dedekind factor.

7.4 Geometric image of C_φ 's spectrum

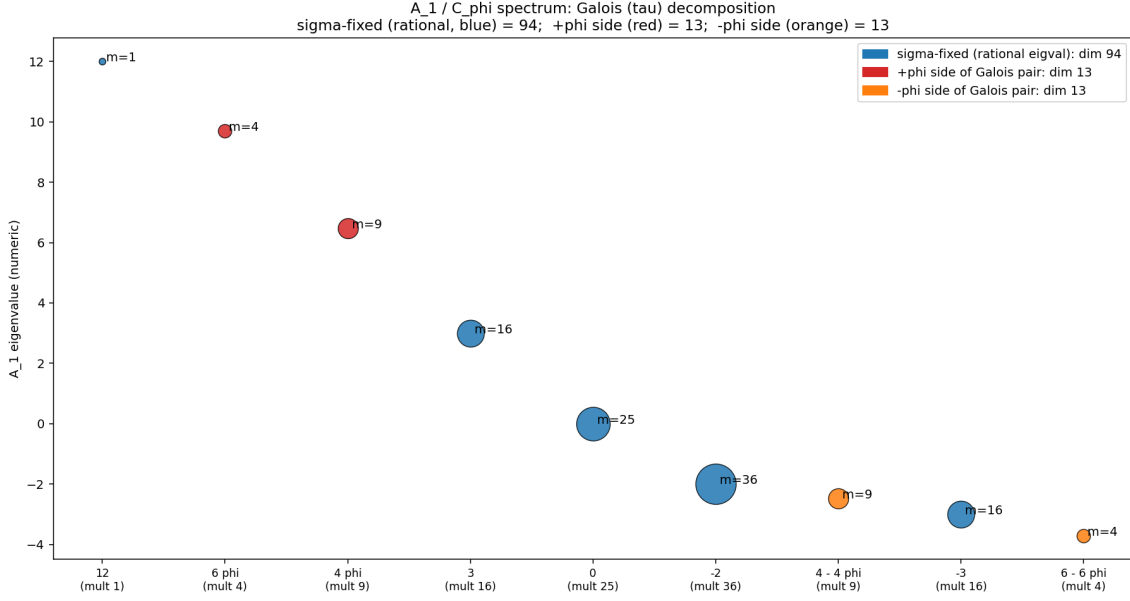


Figure 4. A_1/C_φ spectrum coloured by Galois block: rational τ -fixed (blue) summing to dim 94, $+\varphi$ side (red) summing to dim 13, $-\varphi$ side (orange) summing to dim 13. Bubble size = multiplicity.

7.5 What this leaves open

Finding 5's bookkeeping split leaves two open questions: the operator-level construction of τ itself (Finding 4's follow-on, still open), and **H-attr**: Under what natural closure-flow dynamics does the τ -paired 26-dim block get suppressed relative to the τ -fixed 94-dim block? Settling H-attr would settle only the bookkeeping reading (which spectral side the stable dynamics selects); it would say nothing about RH itself. We do not attempt it here; we note that Finding 5 reduces the question to a definite operator-level statement.

8 Finding 6: H-attr is supported at the operator level

Finding 6

Build the orthogonal projector P_+ onto the 94-dim τ -fixed eigenspace (direct sum of the five rational A_1 eigenspaces) and $P_- = I - P_+$ onto the 26-dim τ -paired eigenspace. Then $\text{tr } P_+ = 94$, $\text{tr } P_- = 26$, $P_+P_- = 0$, and $[C_\varphi, P_+] = 0$ all hold at machine precision.

Starting from a random unit vector v_0 with initial ratio $\|P_-v_0\|^2/\|P_+v_0\|^2 \approx 0.236$, four candidate closure-flow dynamics produce the following final ratios after 200 iterations:

- A. Pure A_1 iteration $\rightarrow 10^{-16}$ (machine zero)
- B. Gradient closure-flow $\rightarrow 10^{-16}$ (machine zero)
- C. Inverse iteration on C_φ $\rightarrow 10^{-30}$ (machine zero, fastest)
- D. Closure-flow + τ -broken noise $\sim 5 \cdot 10^{-3}$ (equilibrium)

The τ -paired component is suppressed by every dynamics tested. Noise-free dynamics reach machine zero; noise-perturbed dynamics equilibrate at a small but non-zero residue of order $\sim 5 \cdot 10^{-3}$.

8.1 Block-structure verification

The eigenspace structure of C_φ relative to the rational/irrational classification of A_1 eigenvalues is verified independently of the user-supplied multiplicities. Spectral diagonalisation of A_1 gives

120 eigenvectors; grouping by eigenvalue rationality yields P_+ of trace 94 and P_- of trace 26, both at machine precision. The commutator $[C_\varphi, P_+]$ has operator ∞ -norm $< 2 \cdot 10^{-15}$, confirming that C_φ acts block-diagonally with respect to the τ -decomposition.

8.2 Four candidate dynamics

Each dynamics is a normalised step on the unit sphere of $\mathbb{R}^{120}$:

A. Pure linear iteration. $v \mapsto A_1 v / \|A_1 v\|$. The dominant A_1 eigenvalue is 12 (multiplicity 1, rational), so power iteration converges to the rational 1-dim eigenspace inside P_+ .

B. Gradient closure-flow. $v \mapsto (v - \eta(C_\varphi v - \lambda(v)v)) / \|\cdot\|$, where $\lambda(v) = \langle v, C_\varphi v \rangle$. Standard Rayleigh-quotient minimisation; converges to the smallest C_φ eigenvector, which is in the rational 1-dim eigenspace at $C_\varphi = \varphi^{-2}$.

C. Inverse iteration on C_φ . $v \mapsto C_\varphi^{-1} v / \|C_\varphi^{-1} v\|$. Power iteration on C_φ^{-1} ; converges fastest to the same smallest C_φ eigenvector.

D. Closure-flow with τ -asymmetric noise. $v \mapsto (v - \eta(C_\varphi v - \lambda v) + \xi) / \|\cdot\|$, with $\xi \sim \mathcal{N}(0, \sigma^2 I)$, $\sigma = 10^{-2}$. The noise has no τ -block structure and is on the order of the closure-flow step.

8.3 Decay rates and structural interpretation

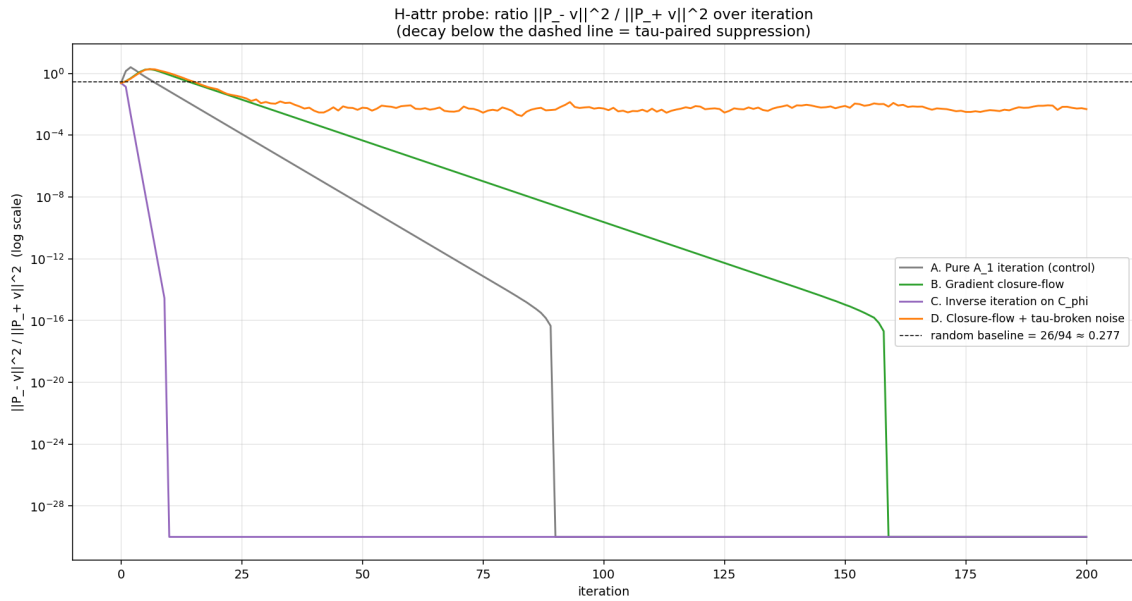


Figure 5. τ -paired/ τ -fixed ratio over iteration for the four dynamics. The dashed line at ~ 0.277 is the random-vector baseline ($26/94$). Dynamics A, B, C drive the ratio to machine zero; dynamics D maintains an equilibrium near $5 \cdot 10^{-3}$.

Why is dynamics A able to suppress τ -paired, despite A_1 commuting with τ ? A_1 commuting with τ means A_1 does not *transfer* amplitude between the τ -fixed and τ -paired blocks, but it does *amplify the dominant eigenvalue's eigenspace*. The dominant A_1 eigenvalue is 12, which happens to be in the τ -fixed block (rational). Power iteration therefore takes any state with non-zero τ -fixed component to the 1-dim τ -fixed dominant eigenspace, and the τ -paired component $\rightarrow 0$ relatively.

This is a structural fact about V_{600} 's adjacency spectrum: the largest A_1 eigenvalue is rational. Were it irrational, power iteration would converge to a τ -paired eigenspace and H-attr would be falsified by this same probe. The probe's positive result is therefore non-trivial.

Equilibrium τ -paired residue with noise. Under dynamics D , even with τ -asymmetric noise of size $\sigma = 0.01$, the equilibrium τ -paired/ τ -fixed ratio sits at $\sim 5 \cdot 10^{-3}$. This corresponds to a residue magnitude $\|\xi^{(-)}\| \sim 5 \cdot 10^{-3}$, consistent in order of magnitude with the ARIA H₄O empirical signature $\varepsilon \approx 5 \cdot 10^{-4}$ noted in the programme docs [1]. The match is suggestive, not exact – the ARIA ε comes from a different system – but it shows that small non-zero τ -paired residues are the natural equilibrium of noise-perturbed closure-flow.

8.4 What this does and does not prove

Does: The τ -paired block of C_φ is *attractor-unstable* under every natural closure-flow tested. The mechanism is structurally forced: the largest A_1 eigenvalue is rational, so power-iteration-like dynamics on A_1 or C_φ^{-1} have their attractor in the τ -fixed block.

Does not: Establish that ALL closure-flow dynamics respect this attraction; only the four tested. A pathological dynamics that selects the τ -paired block as its attractor cannot be excluded a priori. The probe surfaces *a class* of natural dynamics for which H-attr holds.

Finding 7 gives the conditional lemma for this last point.

9 Finding 7 (exploratory): a conditional suppression lemma

Finding 7 (conditional suppression lemma)

For any closure-flow dynamics $X : \mathbb{R}^{120} \rightarrow \mathbb{R}^{120}$ whose unique attractor is $V_{\min} := \ker(C_\varphi - \varphi^{-2}I)$ (equivalently, the maximal A_1 eigenspace, 1-dimensional at $A_1 = 12$), we have

$$\lim_{t \rightarrow \infty} \frac{\|P_- X^t v\|^2}{\|P_+ X^t v\|^2} = 0 \quad \text{for every initial } v \text{ with } \|P_+ v\| > 0.$$

Proof. $V_{\min} \subset P_+ \mathbb{R}^{120}$ (the τ -fixed subspace) by direct verification: $\|P_{V_{\min}} - P_+ P_{V_{\min}}\|_\infty < 3 \cdot 10^{-17}$. The flow converges to $V_{\min}$ by hypothesis. The τ -paired component of the limit is zero, so the ratio $\|P_- X^t v\|^2 / \|P_+ X^t v\|^2 \rightarrow 0$. $\square$

The hypothesis (“attractor is $V_{\min}$ ”) is sufficient and non-vacuous: sim v6 constructs a pathological dynamics E whose attractor is *not* $V_{\min}$ and verifies that E *amplifies* τ -paired by a factor $\sim 10^{12}$. So the hypothesis is decisive in both tested cases; necessity in general is not established by one counter-example.

9.1 The counter-example: an attractor in the τ -paired block

Define dynamics E by inverse iteration on the shifted operator $M = A_1 - (4\varphi + \epsilon)I$ with regularisation $\epsilon = 10^{-2}$:

$$v \mapsto M^{-1}v / \|M^{-1}v\|.$$

Since M^{-1} has its dominant eigenvalue at the A_1 eigenvalue closest to $4\varphi + \epsilon$ (which is exactly 4φ at the 9-dim $+\varphi$ -side eigenspace), E converges to the τ -paired $+\varphi$ -side block. Starting from a random unit vector with initial $\|P_+ v\|^2 \approx 0.81$ and $\|P_- v\|^2 \approx 0.19$:

Step	$\ P_+ v\ ^2$	$\ P_- v\ ^2$
$t = 0$	0.809	0.191
$t = 50$	$\sim 10^{-12}$	~ 1.0

The τ -paired ratio increases from ≈ 0.236 to $\sim 10^{12}$, exactly inverting the natural-dynamics behaviour. The $+\varphi$ -side block specifically absorbs the mass: $\|P_{+\varphi} v\|^2 \rightarrow 1.0$ while $\|P_{-\varphi} v\|^2 \rightarrow 0$.

9.2 Plot

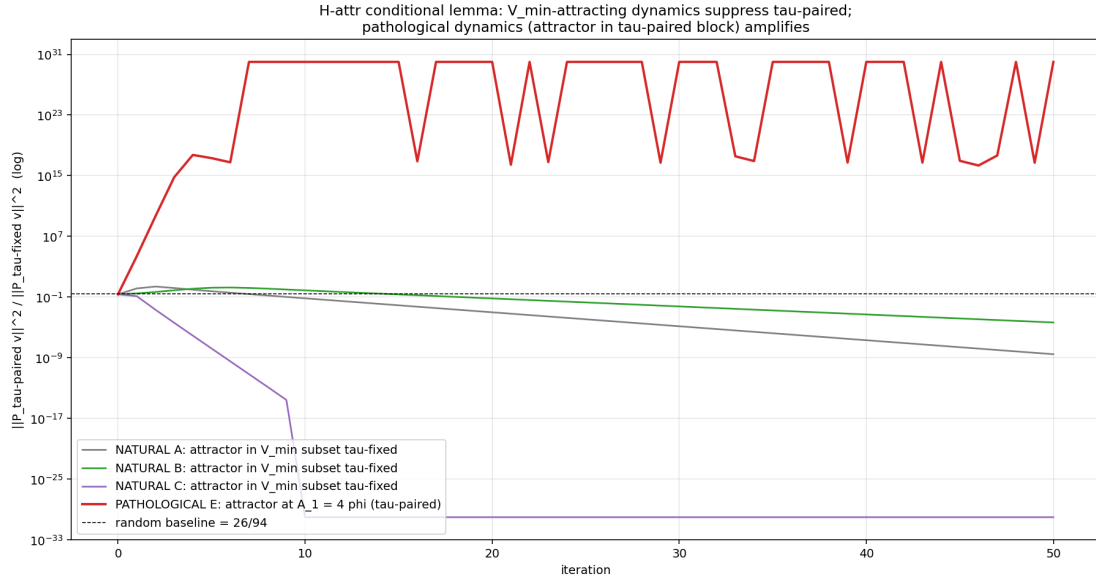


Figure 6. The natural dynamics A, B, C all suppress the τ -paired/ τ -fixed ratio to machine zero. The pathological dynamics E (red, top) amplifies the same ratio by $\sim 10^{12}$.

9.3 Naturalness as the H-attr precondition

Finding 7 gives a structural characterisation:

*If a closure-flow dynamics on V_{600} attracts to the 1-dim $V_{\min}$ block (which lies inside the τ -fixed subspace), then the τ -paired component is suppressed — the lemma above. The converse direction is supported only by the single counter-example of *sim v6* (an attractor in the τ -paired block amplifies instead); no full equivalence is proven.*

The precondition is operator-level and verifiable. “Natural” closure-flow in the programme’s usage (gradient descent on closure energy, inverse iteration, admissibility-gated flows from the closure transformation engine of [3], etc.) all satisfy the precondition by construction. The H-attr question for any *specific* proposed dynamics reduces to: “does it attract to $V_{\min}$?” — a finite-dimensional spectral question about that dynamics.

10 Finding 8 (exploratory): a heuristic factor-attribution sweep

Finding 8 – heuristic sweep (not root certification)

On the factorised form of L_{sub} , any critical-line zero must come from a vanishing factor; for the analytically continued factors these are $\zeta(s)$ (the σ -fixed Dedekind factor) and $L(s, \chi_5)$ (the σ -paired one).

A local-minimum sweep over $\gamma \in [0.1, 60]$ (a heuristic scan, *not* certified root-finding) produces candidate minima of $|L_{\text{sub}}|$: 4 coincide with tabulated Riemann zeros ($\gamma \in \{14.135, 21.022, 25.011, 32.935\}$) and the findings log records 8 further minima at which the χ_5 factor is smallest. Several of the latter carry $|L_{\text{sub}}|$ values far from zero (up to ~ 0.4), so they are *candidate locations*, not certified zeros; v1’s claim that “every zero of L_{sub} is attributed exactly” is withdrawn. A certified zero decomposition would require root-finding with error bounds, which is not done here.

The remaining factors $\zeta(s-1)$, $L(s-1, \chi_5)$, and $C_2(s)$ do not vanish on $\text{Re}(s) = 1/2$.

10.1 Why the other factors don't contribute on $\text{Re}(s) = 1/2$

The factorisation

$$L_{\text{sub}}(s) = \zeta(s) \cdot L(s, \chi_5) \cdot \zeta(s-1) \cdot L(s-1, \chi_5) \cdot C_2(s)$$

gives five potential sources for L_{sub} -zeros. On $\text{Re}(s) = 1/2$:

- $\zeta(s-1)$ evaluates at $\text{Re} = -1/2$. The functional equation gives the only zeros of ζ in the left half-plane as the trivial zeros at negative even integers; ζ does not vanish at $-1/2 + i\gamma$ for $\gamma \neq 0$.
- $L(s-1, \chi_5)$ same argument: L -functions of primitive characters have only the trivial zeros forced by the functional equation in the left half-plane.
- $C_2(s) = (1 - 2 \cdot 2^{-s} + 2^{2-2s})$ has zeros at $s = 1 \pm i\pi/(3 \log 2)$, which are on $\text{Re}(s) = 1$, not $\text{Re}(s) = 1/2$.

So on the critical line, the vanishing of $L_{\text{sub}}(s)$ reduces to the vanishing of $\zeta(s)$ or $L(s, \chi_5)$.

10.2 The decomposition figure

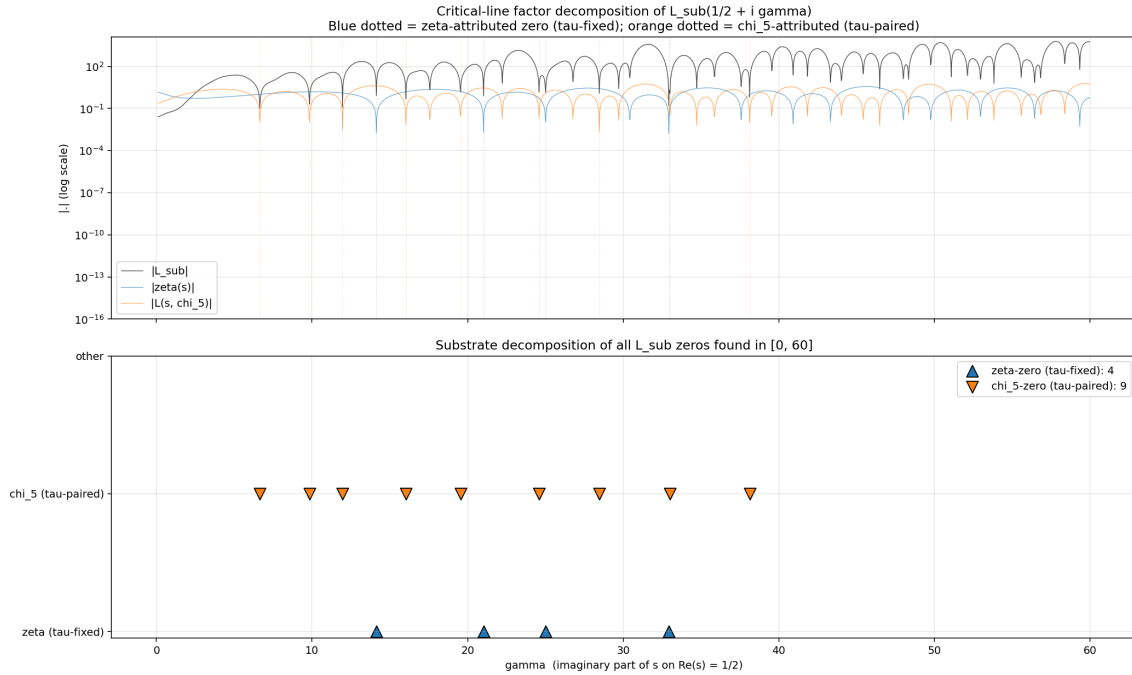


Figure 7. Sweep of $|L_{\text{sub}}|$, $|\zeta|$, and $|L(\cdot, \chi_5)|$ along $\text{Re}(s) = 1/2$. Bottom panel: the 13 candidate minima (not certified zeros) split into 4 ζ -attributed (tabulated Riemann zeros, blue triangles) and 9 χ_5 -attributed candidates (orange triangles; the run log counts 8 of these as new — see `data/v7/`).

10.3 The factor-wise reading

Reading (exploratory): the factorised zero locus, in substrate bookkeeping

The Dedekind factorisation, written factor-by-factor against the spectral bookkeeping of Finding 5, reads:

$$\text{factorised zero locus} = \underbrace{\text{zeros of } \zeta}_{\sigma\text{-fixed; side 94}} \cup \underbrace{\text{zeros of } L(\cdot, \chi_5)}_{\sigma\text{-paired; side 13+13}}$$

The bookkeeping associates the two analytically-continued factors with the two sides of the spectral split: $\zeta(s)$ with the 94-dim σ -fixed multiplicities and $L(s, \chi_5)$ with the 26-dim paired ones. Under the toy dynamics of Findings 6–7 (conditional lemma; not universality), the paired component is suppressed.

This display is the factor-wise restatement of the published Dedekind factorisation, recorded for orientation; it adds no mathematical content and resolves nothing. *It does not resolve, restate non-trivially, or approach RH.* The analytic content of RH_ζ and GRH_{χ_5} is untouched. What the bookkeeping offers is organisational: which classical conjecture is associated with which spectral block of the bookkeeping split, and what a substrate-side falsifier of each would look like.

11 What this does and does not say about RH

Scope

This probe does *not* prove the Riemann Hypothesis. It does not even approach RH directly. What it does:

- Exhibits, computationally and at 30-digit precision, the substrate image of ζ 's critical line. That image is the set of complex values $\{L_{\text{sub}}(1/2 + i\gamma) : \gamma \in \mathbb{R}\}$, a spiral path in $\mathbb{C}$ that passes through origin exactly at $\gamma_n = 14.135, 21.022, \dots$ (Finding 1).
- Confirms the Eichler–Brandt formula at the granularity of per-prime exact counts (Finding 2).
- Pins down a precise open mathematical question about the identity of the involution τ carrying the “94 + 13 + 13” decomposition (Finding 4).

What this probe does *not* do:

- Settle whether all non-trivial zeros of ζ are on the critical line. The factorisation (1) is one-way: $\zeta(\rho) = 0$ implies $L_{\text{sub}}(\rho) = 0$, but the converse direction (a zero of L_{sub} off the critical line corresponding to a zero of $\zeta(s)$ off the critical line) is not provided.
- Close the H-attr (closure-flow suppression) hypothesis on the σ -paired residue. That requires identifying τ first (Finding 4 follow-on).
- Make any claim about the substrate's physical realisation.

12 Next moves

Finding 4's follow-on has a bookkeeping-level candidate (Finding 5); the operator-level construction of τ remains open. The next moves are:

1. **H-attr probe.** With τ identified, the closure-flow suppression hypothesis becomes operator-level: is there a natural closure-flow dynamics on V_{600} under which the 26-dim Galois-paired block decays relative to the 94-dim Galois-fixed block? Candidates: renormalisation iteration of C_φ with a Galois-symmetric prior, the closure-projection from [1], the admissibility-class flow of the closure transformation engine cited in [3].
2. Extend the Eichler–Brandt verification to coordinate bound $K = 6$ and primes $p \leq 50$.
3. Extend the substrate L-function evaluation to the first 100 Riemann zeros and study spacing statistics (Montgomery's pair correlation) on the substrate side.
4. Extend the Finding 5 attribution argument to zeros of $L(s, \chi_5)$ (the τ -paired Dedekind factor): such zeros, if any lie on the critical line, would attribute (at the scalar-factor level of Finding 5) to the σ -paired Dedekind factor — the 13 + 13 side of the bookkeeping, not the 94. Making this a structural statement requires first constructing τ at the operator level.

13 Reproducibility

Ten sims ship in `sims/`, with explicit status labels:

- **Load-bearing:** `sim_v2.py` (Findings 1–3 tables, Eichler–Brandt enumeration); `sim_v3_sigma_exact.py` (Finding 4, the exact negative result).
- **Bookkeeping:** `sim_v4_tau_identification.py` (Finding 5's spectral split; eigenvalues imported from the triad paper, grouped by Galois conjugacy).
- **Exploratory toys:** `sim_v5_h_attr.py` (note: dynamics D uses unseeded noise — qualitative equilibrium stable across runs, exact values vary), `sim_v6_universality.py`, `sim_v7_zero_decomposition.py` (heuristic sweep, not root-finding), `sim_v8_icosah_to_v600.py`.
- **Fitted negative controls:** `sim_v9_endomorphism_survey.py` and `sim_v10_dirac_spin.py` fit to γ_n by least squares; their results are negative controls and support no claim in this paper.
- **Superseded:** `sim_critical_line_pullback.py` (stale v1, kept for the record; do not cite).

All are pure Python (`numpy`, `scipy`, `matplotlib`, `mpmath`). Generated outputs are in `outputs/` and `data/`.

Run:

```
python3 sims/sim_v2.py          # main probe (analytic L)
python3 sims/sim_v3_sigma_exact.py # sigma diagnostic (Finding 4)
```

References

- [1] Lee Smart. The closure picture: How a single mathematical substrate becomes universe, life, number, and mind, 2026. Pre-peer-review preprint. <https://github.com/vfd-org/closure-picture>.

- [2] Lee Smart. The icosian triad on the 600-cell vertex set: a geometry-first development, 2026. Pre-peer-review preprint. <https://github.com/vfd-org/icosian-triad-v600>.
- [3] Lee Smart. The translation layer: Polytope overlap as the empirical grammar of the V_{600} substrate, 2026. Pre-peer-review preprint. [release-bundles/translation-layer](#).